

HexaCraft: An Interactive Hexagonal Voxel World

A Procedurally Generated Terrain Exploration System using OpenGL

Proposed By - Md. Faysal Mahmud (2007068), Niloy Day (2007086)

Introduction

HexaCraft reimagines the voxel world paradigm by implementing a hexagonal block system, where each “hexel” (hexagonal prism) organizes to form organic, honeycomb-like terrain structures. Additionally, the project explores procedural terrain generation, dynamic camera systems, and interactive block manipulation within a fully navigable 3D environment.

Unlike static demonstrations, HexaCraft emphasizes player agency, users can explore vast landscapes, interact with the environment, and experience multiple viewing perspectives from first-person immersion to strategic bird’s-eye overviews.

Objectives

The primary objectives of this project are:

- **To implement a hexagonal voxel rendering system** with efficient mesh generation and culling techniques for hexagonal prisms.
- **To develop procedural terrain generation using noise-based algorithms** that create natural, varied landscapes.
- **To create a robust multi-perspective camera system** allowing seamless transitions between first-person, third-person, orthogonal, and bird’s-eye views.
- **To implement player physics** including collision detection, gravity simulation, and basic movement mechanics adapted to hexagonal geometry.
- **To explore advanced lighting and shading techniques** comparing Gouraud and Phong shading models with per-block texture mapping.
- **To design an interactive environment** where users can manipulate terrain and observe dynamic objects responding to physics.

Technical Implementation Idea

The project will be built using C++ and OpenGL specifications (utilizing GLFW for window management and input handling along with GLM for maths).

Core Components

- **Hexagonal Geometry:** Each hexel is constructed from 8 faces (6 rectangular sides + 2 hexagonal caps). Vertices are generated programmatically to ensure proper tessellation.
- **Chunk-Based World Management:** The infinite world is divided into manageable chunks, loaded and unloaded dynamically based on player position to maintain performance.
- **Transformation Pipeline:** Use of model-view-projection matrices to handle camera transformations, object positioning, and view mode transitions.
- **Texture:** All block textures (grass, sand, stone, wood, etc.) are stored in a single texture atlas, with UV coordinates mapped appropriately per hexagonal face.
- **Collision System:** Custom hexagonal AABB (Axis-Aligned Bounding Box) collision detection adapted for 6-sided prism geometry.

Dynamic Interactive Objects

The simulation responds dynamically to user input and features multiple interactive systems:

1. The Player Avatar

- **Movement:** WASD controls for forward/backward/strafe movement with smooth acceleration.
- **Jumping:** Space bar for vertical movement with gravity-based arc trajectory.
- **Head Movement:** Mouse control for looking around in first-person mode.
- **Block Interaction:** Left-click to break blocks, right-click to place blocks at targeted positions.

2. Dynamic Camera System (View Mode Switching)

- **First-Person View:** Camera bound to player's head position.

- **Third-Person View:** Camera positioned behind and above the player, showing character.
- **Orthogonal Projection:** Top-down 2D view for strategic navigation and terrain overview.
- **Bird's Eye View:** Elevated isometric perspective showing larger terrain sections and spatial relationships.

3. Interactive Block System

- **Block Mining:** Target a block with the crosshair and destroy it, updating the mesh in real-time.
- **Block Placement:** Select a block type from inventory and place it in the world, with collision checks to prevent invalid placements.
- **Physics Response:** Blocks affected by gravity and other blocks collision.

4. Day-Night Cycle Control

- **Manual Time Setting:** Jump to specific times of day (dawn, noon, dusk, night) for observing lightning changes.

5. Environmental Interactions

- **Falling Blocks:** Unsupported sand and gravel blocks trigger physics simulations and fall realistically.

Objects with Complex Shapes

Beyond standard hexagonal prism blocks, the scene includes geometrically complex objects:

1. The Player Character

Unlike simple blocks, the player is a composite hierarchical model:

- **Body Structure:** Torso, head, arms, and legs constructed from scaled hexagonal prisms and cylinders.
- **Joints:** Rotation points at shoulders, elbows, hips, and knees for walking animations.

- **Hand-held Items:** Tools or blocks held in hand, dynamically swapped based on inventory selection.

2. Trees

- **Tree Trunks:** Vertical stacks of slightly varied hexagonal blocks creating organic-looking bark texture.
- **Canopy:** Irregular cloud of leaf blocks using randomized placement to avoid perfect spherical shapes. Some trees feature voxelized branching structures.
- **Small Plants:** Flowers, mushrooms, and grass tufts.

3. Hexagonal Vehicle (MineCar)

A drivable transportation object:

- **Chassis:** Composite structure of elongated hexagonal prisms forming a vehicle frame.
- **Wheels:** Cylindrical or disk-shaped components that rotate based on movement speed.
- **Cockpit:** Semi-transparent hexagonal dome for the driver's seat.
- **Animation:** Wheels rotate, suspension compresses on uneven terrain, and particles emit from thrusters or wheels.

4. Animated Entities

- **Flying Creatures:** Simple voxel-based birds or insects following spline paths through the environment.
- **Resource Indicators:** Floating icons or effects marking mineable resources or points of interest.

Conclusion

HexaCraft represents an exploration of alternative voxel geometries in interactive 3D environments. By replacing the traditional cubic block with hexagonal prisms, the project challenges conventional approaches to terrain generation, collision detection, and world representation. The hexagonal tessellation offers unique aesthetic qualities—evoking natural patterns found in honeycombs, columnar basalt formations, and crystalline structures.

Proposed Visuals



Day and Night Scene



Camera Modes and Complex Objects